Alessandro Puglisi

PhD Candidate in Marine Robotics · Underwater Perception

University of Montpellier, LIRMM RSM, France

(+39) 3762501607 | alessandro.puglisi@lirmm.fr | Website | GitHub | LinkedIn

Skills

Programming	C++, Python, MATLAB, C, LaTeX
Software	Linux, git/GitHub, ROS 1/2 (Robot Operating System), Stonefish, Gazebo, Simulink
Libraries and Toolboxes	opencv, matplotlib, pytorch, apriltag, colmap, Nav2
Engineering	Robotics, perception and state estimation, computer vision, deep learning, systems modeling

Experience

MBARI - CoMPAS Lab
Moss Landing, United States

Underwater Robotics Perception Summer Intern — *Jun. 2026 - Aug. 2026*

- Investigated methods for generating high-resolution, semantically labeled 3D representations from ROV imagery to support underwater scene understanding and scientific research.
- Developed and evaluated computer-vision workflows for underwater image enhancement, semantic segmentation, and 3D reconstruction of benthic environments, using Gaussian Splatting.
- Participated in field data collection aboard the R/V Rachel Carson using the scientific camera of ROV Ventana.
- Presented project methodology and results to an audience of ocean scientists (recording link).

INESCTEC - Robotair
Porto, Portugal (Remote)

Robotics Python Developer Summer Intern — *Jul. 2025 - Sept. 2025*

- Developed new python features for Robotair, a platform to simplify robotics software development and deployment.
- Handled version control best practice in a software project involving four different teams.

Space Robotics Lab (SRL)
Sendai, Japan

Research Student — *Apr. 2024 - Aug. 2024*

- Conducted 500+ hours of research in Tohoku University's largest robotics lab, which comprises 80+ members across 5+ teams, under the supervision of Yoshida Kazuya Sensei.
- Researched methods for robustly and efficiently transforming a prototype multi-modal limbed climbing robot, equipped with a transformable gripper-wheel module.
- Integrated the new module in high and low-level control software, with a ROS2 architecture written in C++ and python.
- Tested the updated software and transition algorithm on the physical system, achieving system stability and eliminating steady-state error in motor control.

Projects

Simulation of MBARI's Underwater Vehicles

Personal Project — *Jan. 2026 - Present*

- Defined 3D models of MOLA AUV and ROV Doc Rickets (using blender). Integrated thrusters actuation, teleoperation, and sensing, based on vehcicle's official specifications and MBARI's research papers.
- Tested state of the art computer vision and robotics algorithms on custom modeled world (Kalman filtering, Structure from Motion Colmap, 3D Gaussian Splatting gsplat, apriltag, ORB-SLAM3).
- New simulation projects are under development. e.g. Lightweight manipulator for MOLA, deep sea animals identification and tracking, autonomous coral reef restoration farm navigation.

ROS2 Driver for Blueprint Oculus M3000d Forward Looking (Imaging) Sonar

PhD Experiment set-up — *Dec.2025 - Present*

- Investigated working principle of acoustic sensing, related software interface, and used data structures.
- Updated an incompatible sensor drivers using the proprietary C++ software, for the M3000d sensor model.
- Integrated acoustic data pre-processing and explored possible state estimation approaches (e.g. odometry via acoustic image optical flow or from sonar's internal AHRS).

Modeling, Control, and Dexterity Analysis of a Multi-Modal Limbed Robot

M.S. Thesis, Follow Up of the Space Robotics Lab Research *Sept. 2024 - Mar.2025*

- Modeled the kinematics and dynamics of a transformable gripper-wheel limbed robot in MATLAB.
- Researched performance indices for the maneuverability of limbed systems, deriving a novel theory of dexterity.
- Evaluated the reliability of the new performance criteria, with a nature-inspired example, based on grasshopper insect kinematics.
- Designed a new robot controller in MATLAB and Simulink, Implemented and simulated both in wheeled and legged locomotion mode, with ROS 2 (C++) and Gazebo.

Mobile Robot Odometry, Mapping and Way-points Navigation

Perception, Localization, and Mapping for Mobile Robots *Apr. 2023 - July 2023*

- Implemented odometry (position) estimation for an autonomous shuttle, with ROS and C++.
- Mapped an indoor environment using ROS Laser Scan and 3D Lidar data, with the slam_toolbox.
- Denoised the map, and performed autonomous navigation from csv waypoints, localizing the system with the AMCL algorithm (Adaptive Monte Carlo Localization) and the Navigation stack for path generation and tracking.

Education

University of Montpellier, RSM LIRMM *Montpellier, France*

PhD Candidate in Underwater Robotics Perception *Oct. 2025 - 2028*

- **PhD Thesis**: *Mapping and monitoring of the surface condition and biological colonization of submerged parts of offshore wind turbines using acoustic imaging*.
- **Advisors**: Vincent Creuze, Juliette Drupt, Frederic Comby
- **Topics**: Semantic opti-acoustic simultaneous localization and mapping, sensor fusion, deep learning.

Max Planck Institute of Animal Behavior *Konstanz, Germany*

WildDrone Summer School *Sept. 2025*

- **Program**: Intensive week of talks and workshops from leading multidisciplinary researchers in the field of robotics and biology. Discussing relevant research projects, tools, and best practice to fill the gap between the two domains.
- **Topics**: Robotics, environmental monitoring, eDNA, conservation biology, animal behavior, remote sensing.

Politecnico di Milano *Milano, Italy*

M.S. in Automation and Control Engineering *Sept. 2022 - Apr. 2025*

- **MSc Thesis**: Motion Control and Manipulability Analysis of LIMBERO-GRIEEL: a Multimodal Limbed Robot for Unstructured Environments.
- **Relevant Courses**: Perception Localization and Mapping, Control of Industrial and Mobile Robots, Networked Control, Data Driven Control, Model Identification and Data Analysis, Dynamics of Mechanical Systems.

Tohoku University *Sendai, Japan*

Exchange Student at Space Robotics Lab *Apr. 2024 - Aug. 2024*

- **Courses** : System Control Engineering, Laboratory Experiments.

Politecnico di Milano *Milano, Italy*

B.S. in Automation and Control Engineering *Sept. 2019 - Sept. 2022*

- **Relevant Courses**: Fundamentals of Robotics, Informatics, Linear Algebra and Geometry, Probability, Calculus 1 and 2, Electronics, Measurement and Instrumentation, Modeling of Mechanical Systems.

Extra Activities and Certificates

Volunteering

Ailerons (Joined Oct.2025): Non-profit multidisciplinary scientific association.

Primary objectives of study, protect and educate about sharks and rays in the Mediterranean.

Formation

- **Ocean and Conservation:** Scuba Diving SSI Advanced Adventurer, FESSMM Niveau 2, Nitrox, Dry Suit, SSI Blue Oceans, Coral Reef Restoration, Remote Sensing and Mapping for Coral Reef Conservation, IUCN Red List (conservationtraining.org)
- **Software Tools:** ROS2 Basics and Intermediate, ROS2 Nav2, Complete Computer Vision Bootcamp With PyTorch & Tensorflow (Udemy), OpenCV, TensorFlow-Keras, PyTorch Bootcamp (OpenCV University)